

Records identified from databases:

Scopus (n = 690)
ACM Digital Library (n = 184)
IEEE Xplore (n = 16)
Web of Science (n = 11)
Springer Link (n = 44)
SBC OpenLib (n = 23)

Records removed before screening:

Published before 2021
(n = 120)
Non-Computer Science thematic areas
(n = 259)
Non-peer-reviewed publications
(n = 229)
Other languages
(n = 1)
Duplicates
(n = 19)

Records screened

(n = 340)

Records excluded:

Non-primary research and non-peer-reviewed publications
(n = 20)
Contexts other than higher education
(n = 67)
Studies where student academic health prediction is not primary objective or outcome
(n = 139)
Studies where student academic health assessment is conducted exclusively through manual or non-automated methods
(n = 34)

Reports sought for retrieval

(n = 80)

Reports not retrieved:

Full text unavailable (n = 13)

Reports assessed for eligibility

(n = 67)

Reports excluded:

Quality score below 3.75 (n = 14)

Studies included in review

(n = 53)

Studies per teaching environment:

In-person (n = 24)
Hybrid (n = 10)
Online (n = 19)